Machine Learning-Based Classification of Breast Cancer Molecular Subtypes Using Gene Expression Profiles: A Comparative Study with Functional Enrichment Analysis

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Abstract

Breast cancer is one of the most common cancers worldwide and is classified into different molecular subtypes based on gene expression patterns. Since these subtypes differ in their biological characteristics and response to treatment, accurate classification is important for early diagnosis and personalized therapy. In this study, machine learning techniques were used to classify breast cancer molecular subtypes using gene expression data from the publicly available GSE45827 dataset obtained from the Gene Expression Omnibus (GEO) database. After preprocessing, the final dataset consisted of 141 tissue samples belonging to Basal, HER2-positive, Luminal A, Luminal B, and Normal breast tissue groups.

The gene expression data were pre-processed by removing non-tissue samples and extracting subtype labels from the sample metadata. ANOVA F-test was used to select the most informative genes, and Principal Component Analysis (PCA) was performed to visualize the distribution of the selected samples. Four machine learning models—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were trained using an 80:20 train-test split. To obtain a more reliable estimate of model performance, 10-fold stratified cross-validation was also carried out. The models were evaluated using accuracy, confusion matrices, ROC curves, and AUC scores.

Among the evaluated models, Logistic Regression and SVM achieved a test accuracy of 100%, while Random Forest and XGBoost achieved accuracies of 96.55%. The cross-validation results showed consistent performance across all models, with mean accuracies ranging from 90.12% to 96.44%, indicating good generalization on unseen data. The selected genes were further analyzed using Gene Ontology (GO) and KEGG pathway enrichment analysis, which revealed their involvement in several biological processes and pathways associated with breast cancer.

Overall, this study demonstrates that machine learning models, combined with feature selection and functional enrichment analysis, can effectively classify breast cancer molecular subtypes using gene expression data. The results also highlight the usefulness of integrating computational and biological analyses to better understand the molecular mechanisms underlying breast cancer and support future research in precision medicine.

Keywords: *Breast Cancer, Bioinformatics, Gene Expression Analysis, Machine Learning, Classification, Precision Medicine*

Introduction

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide. According to the World Health Organization (WHO), millions of new breast cancer cases are diagnosed every year, making early detection and accurate diagnosis essential for improving patient survival. Breast cancer is not a single disease but consists of several molecular subtypes, including Luminal A, Luminal B, HER2-positive, and Basal (Triple Negative). These subtypes differ in their genetic characteristics, disease progression, prognosis, and response to treatment. Therefore, identifying the *correct molecular subtype is an important step in selecting the most effective treatment strategy for patients.*

With the rapid growth of genomic technologies, gene expression profiling has become an important tool for studying the molecular characteristics of breast cancer. Microarray-based gene expression datasets allow researchers to measure the expression levels of thousands of genes simultaneously, providing valuable information for disease classification and biomarker discovery. However, these datasets are usually high-dimensional, containing thousands of gene expression features but relatively few patient samples. This makes traditional statistical analysis challenging and creates a need for computational methods that can efficiently analyze such complex data.

Machine learning has emerged as a powerful approach for analyzing high-dimensional biological datasets. By learning patterns from gene expression data, machine learning models can classify different breast cancer subtypes with high accuracy and help identify genes that contribute most to the prediction process. In recent years, several algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost have been widely used in cancer classification studies because of their ability to handle complex datasets and generate reliable predictions.

In addition to accurate classification, identifying biologically significant genes is equally important. Feature selection techniques help reduce the dimensionality of gene expression data by selecting only the most informative genes while removing redundant or less relevant features. These selected genes can then be further investigated using functional enrichment analysis, such as Gene Ontology (GO) and Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway analysis, to better understand the biological processes and pathways associated with breast cancer development.

In this study, the publicly available GSE45827 gene expression dataset from the Gene Expression Omnibus (GEO) database was used to classify breast cancer molecular subtypes. After preprocessing the dataset, non-tissue samples were removed, and gene expression data from 141 tissue samples *representing Basal, HER2-positive, Luminal A, Luminal B, and Normal breast tissues were included in the analysis. ANOVA F-test was applied for feature selection, followed by Principal Component Analysis (PCA) for data visualization. Four supervised machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were trained and evaluated using both an 80:20 train-test split and 10-fold stratified cross-validation. Model performance was assessed using accuracy, confusion matrices, ROC curves, and AUC scores. Finally, the selected genes were analyzed using GO and KEGG enrichment analysis to explore their biological significance.*

The main objective of this study was to compare the performance of different machine learning algorithms for breast cancer molecular subtype classification and to investigate the biological relevance of the selected genes. By combining machine learning techniques with functional enrichment analysis, this work demonstrates how computational methods can support breast cancer research and contribute to the growing field of precision medicine.

Literature Review

Breast cancer is one of the most widely studied cancers because of its high prevalence and significant impact on global health.and the complexity of its molecular characteristics. Over the years, researchers have shown that breast cancer is not a single disease but consists of different molecular subtypes, each with its own gene expression pattern, prognosis, and response to treatment. A landmark study by Perou et al. (2000) introduced the molecular classification of breast cancer into Luminal A, Luminal B, HER2-positive, and Basal-like subtypes. This classification has become an important foundation for breast cancer diagnosis and treatment planning.

The development of microarray technology has made it possible to measure the expression levels of thousands of genes in a single experiment. As a result, large-scale gene expression datasets have become publicly available through databases such as the Gene Expression Omnibus (GEO). These datasets have allowed researchers around the world to investigate the molecular basis of breast cancer and develop computational methods for disease classification. The GSE45827 dataset used in this study is one of the publicly available datasets that contains gene expression profiles from different breast cancer subtypes and has been used in several genomic studies.

In recent years, machine learning has become an important tool for analyzing gene expression data. Since these datasets contain thousands of genes but relatively few samples, traditional statistical methods often face challenges in identifying meaningful patterns. Machine learning algorithms are well suited for this type of data because they can learn complex relationships between genes and classify samples with high accuracy. Algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost have been successfully applied in several cancer prediction studies and continue to show promising results.

Another important aspect of gene expression analysis is identifying the genes that contribute most to disease classification. Feature selection techniques help reduce the number of genes by selecting only the most informative ones, making the models more efficient and easier to interpret. In addition, functional enrichment analysis using Gene Ontology (GO) and KEGG pathways helps researchers understand the biological roles of these genes and the pathways in which they are involved. This provides biological meaning to the computational results rather than focusing only on prediction accuracy.

Although many previous studies have reported high classification performance using machine learning, a large number of them mainly focus on developing predictive models. Relatively fewer studies combine machine learning with biological interpretation of the selected genes. In this study, an attempt was made to address both aspects. Gene expression data from the GSE45827 dataset were pre-processed, informative genes were selected using the ANOVA F-test, and four machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were evaluated using both an independent test set and 10-fold stratified cross-validation. The selected genes were further analyzed using GO and KEGG enrichment analysis to better understand their biological significance in breast cancer. By combining predictive modeling with functional analysis, this study provides a more comprehensive approach to breast cancer molecular subtype classification.

Study	Dataset	Methodology	Main Findings
Perou et al. (2000)	Breast tumor samples	Gene expression profiling	Identified intrinsic molecular subtypes of breast cancer.
Gruosso et al. (2016)	GSE45827	Microarray gene expression analysis	Developed a publicly available dataset representing multiple breast cancer molecular subtypes.
Recent machine learning studies	GEO and TCGA datasets	SVM, Random Forest, XGBoost	Demonstrated that machine learning models can accurately classify breast cancer subtypes.
Present Study	GSE45827	ANOVA feature selection, PCA, Logistic Regression, SVM, Random Forest, XGBoost, GO and KEGG enrichment analysis	Compared multiple machine learning models and combined predictive analysis with biological interpretation of selected genes.

The reviewed studies demonstrate that machine learning has become an effective approach for breast cancer subtype classification. However, there is still scope to integrate predictive modeling with biological interpretation of the selected genes. The present study attempts to address this by combining machine learning techniques with functional enrichment analysis using the GSE45827 gene expression dataset.

Materials and Methods

Dataset Description

The gene expression dataset used in this study was obtained from the publicly available Gene Expression Omnibus (GEO) database maintained by the National Center for Biotechnology Information (NCBI). The GSE45827 dataset was selected because it contains gene expression profiles from different breast cancer molecular subtypes and has been widely used in breast cancer research. The expression data were generated using the Affymetrix Human Genome U133 Plus 2.0 microarray platform (GPL570), which measures the expression levels of thousands of genes simultaneously.

The original dataset contained 155 samples representing different breast cancer subtypes and tissue sources. During preprocessing, non-tissue samples and entries without relevant subtype information were removed. The final dataset consisted of 141 breast tissue samples belonging to five classes: Basal, HER2-positive, Luminal A, Luminal B, and Normal breast tissue.

Data Preprocessing

The downloaded gene expression matrix and sample metadata were imported into Python for preprocessing. Sample subtype information was extracted from the metadata, and only tissue samples representing the selected breast cancer subtypes were included in the analysis. Samples with incomplete or irrelevant subtype annotations were excluded to improve the quality of the dataset.

Since gene expression datasets contain a large number of variables, no missing values were observed after preprocessing, and the data were organized into a feature matrix consisting of gene expression values and their corresponding subtype labels. This cleaned dataset was then used for feature selection and model development.

Feature Selection

Gene expression datasets usually contain tens of thousands of genes, many of which contribute little to the classification task. To reduce the dimensionality of the dataset and improve model performance, feature selection was performed using the ANOVA F-test. The ANOVA F-test ranks genes according to their ability to distinguish between different breast cancer subtypes.

To avoid information leakage, feature selection was performed using only the training data within each modeling pipeline before evaluating model performance. The highest-ranked genes were selected and used for model training, reducing computational complexity while retaining the most informative features.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was performed to visualize the distribution of breast cancer samples based on the selected gene expression features. PCA reduces high-dimensional gene expression data into a small number of principal components while preserving most of the variation present in the dataset. The resulting scatter plot provided an overview of sample clustering and the separation among different breast cancer molecular subtypes.

Machine Learning Models

- Four supervised machine learning algorithms were selected for comparison in this study:
- Logistic Regression (LR)
- Support Vector Machine (SVM)
- Random Forest (RF)
- Extreme Gradient Boosting (XGBoost)

The dataset was divided into training and testing sets using an 80:20 stratified split to preserve the class distribution. Each model was trained using the selected gene expression features, and predictions were generated for the testing dataset.

To obtain a more reliable estimate of model performance, 10-fold stratified cross-validation was performed. The mean accuracy and standard deviation obtained from cross-validation were used to evaluate the stability and generalization ability of each model.

Model Evaluation

The performance of each machine learning model was evaluated using several commonly used classification metrics. Test accuracy was calculated to measure the percentage of correctly classified samples. In addition, confusion matrices were generated to examine the classification performance for each breast cancer subtype. Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) were also used to evaluate the discriminative ability of the models.

Cross-validation results were included to assess the consistency of model performance across different subsets of the dataset and to reduce the possibility of overestimating predictive accuracy.

Gene Annotation and Functional Enrichment Analysis

The top-ranked probe IDs obtained after feature selection were converted into gene symbols using the Affymetrix GPL570 platform annotation file. This annotation step enabled the biological interpretation of the selected genes.

To investigate the biological significance of these genes, Gene Ontology (GO) Biological Process enrichment analysis and Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway enrichment analysis were performed using the GSEApy Python package. The significantly enriched biological processes and pathways provided insight into the molecular functions and signaling pathways associated with breast cancer subtype classification.

Results

Model Performance Evaluation

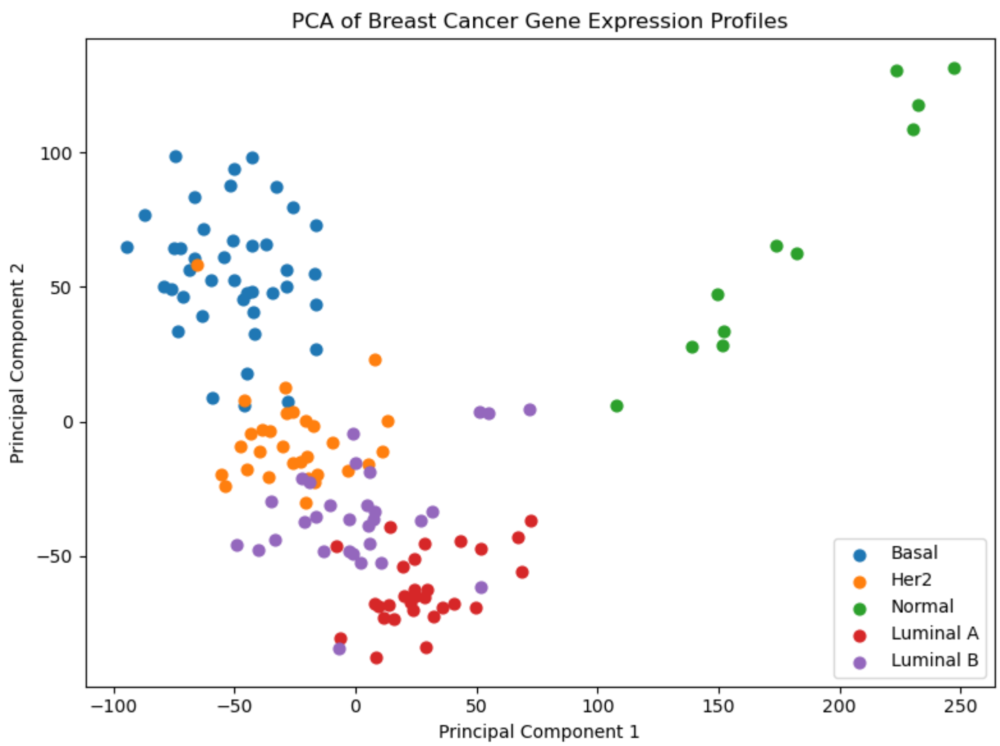
Dataset Summary

After preprocessing, the original GSE45827 dataset was filtered to remove non-tissue samples and entries without relevant molecular subtype information. The final dataset consisted of **141 breast tissue samples** belonging to five molecular subtypes: Basal (41 samples), HER2-positive (30 samples), Luminal B (30 samples), Luminal A (29 samples), and Normal breast tissue (11 samples). The processed dataset contained **29,873 gene expression features**, providing a high-dimensional dataset suitable for machine learning analysis.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was performed to visualize the distribution of breast cancer samples after preprocessing. The PCA plot showed that samples belonging to different molecular subtypes formed distinguishable clusters, although some overlap was observed between closely related subtypes. This indicates that the selected gene expression profiles contain sufficient variation to differentiate breast cancer subtypes while reflecting the biological complexity of the disease.

Figure 1. PCA plot showing the distribution of breast cancer molecular subtypes based on gene expression data.



Performance of Machine Learning Models

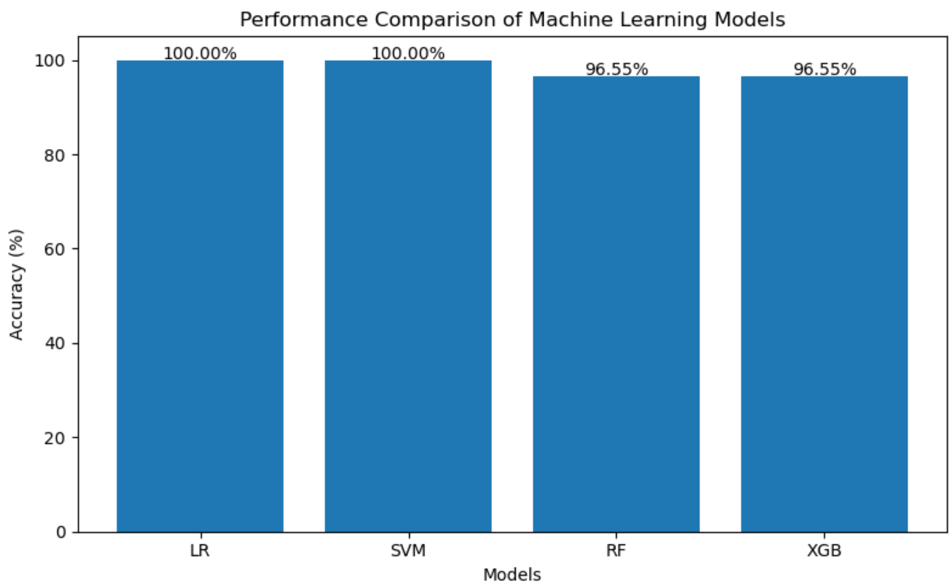
Four supervised machine learning algorithms were evaluated for breast cancer molecular subtype classification. The models were trained using the selected gene expression features and tested using an 80:20 stratified train-test split.

Table 1 summarizes the classification accuracy achieved by each model.

Table 1. Classification Performance of Machine Learning Models

Model	**Test Accuracy (%)**
Logistic Regression	**100.00**
Support Vector Machine	**100.00**
Random Forest	**96.55**
XGBoost	**96.55**

Among the evaluated models, Logistic Regression and Support Vector Machine achieved the highest test accuracy of **100%**, correctly classifying all testing samples. Random Forest and XGBoost also demonstrated strong predictive performance, each achieving an accuracy of **96.55%**.



Cross-Validation Performance

To evaluate the robustness and generalization ability of the models, **10-fold stratified cross-validation** was performed. The average cross-validation accuracy and standard deviation for each model are presented in Table 2.

Table 2. Ten-Fold Cross-Validation Results

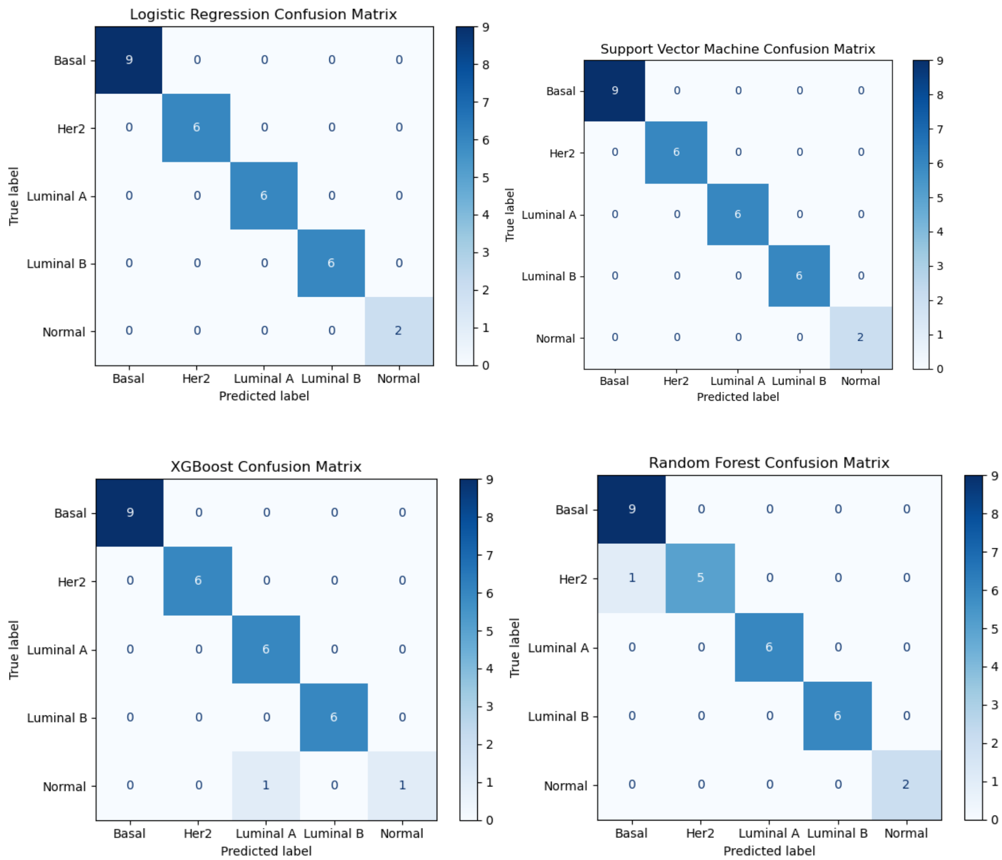
Model	**Mean Accuracy**	**Standard Deviation**
Logistic Regression	94.62%	1.86%
Support Vector Machine	**96.44%**	**1.78%**
Random Forest	94.62%	4.47%
XGBoost	90.12%	5.22%

Although Logistic Regression and SVM achieved perfect test accuracy, the cross-validation results provide a more realistic estimate of model performance on unseen data. SVM achieved the highest average cross-validation accuracy (96.44%) with the lowest variation, indicating stable and consistent performance across different data splits.

Confusion Matrix Analysis

Confusion matrices were generated for all four machine learning models to evaluate their classification performance across individual breast cancer subtypes. Logistic Regression and SVM correctly classified all testing samples without any misclassification. Random Forest and XGBoost produced only a small number of classification errors, demonstrating their ability to accurately distinguish between different molecular subtypes.

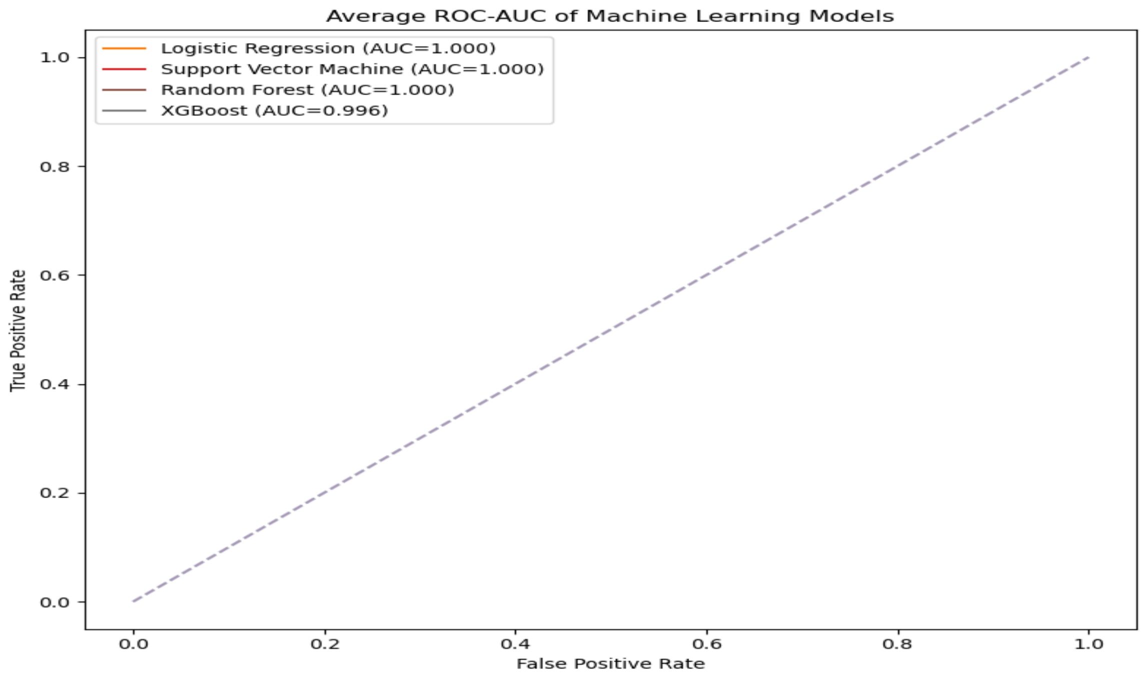
Figure 2. Confusion matrices for Logistic Regression, Support Vector Machine, Random Forest, and XGBoost



Receiver Operating Characteristic (ROC) Analysis

Receiver Operating Characteristic (ROC) curves were generated to evaluate the discriminative performance of each classification model. The ROC curves demonstrated excellent classification ability for all models, with Logistic Regression and Support Vector Machine showing the best overall performance. The high AUC values indicate that the models effectively distinguished among different breast cancer molecular subtypes.

Figure 3. ROC curves comparing the performance of the evaluated machine learning models.



Identification of Important Genes

Feature selection using the ANOVA F-test identified several highly informative genes that contributed to breast cancer subtype classification. Genes such as **ESR1, TPX2, UBE2T, CENPF, CCNB2, UBE2C, RECK, SCARA5,** and **KANK3** were among the top-ranked genes. Many of these genes have previously been associated with breast cancer development, cell proliferation, hormone signaling, and tumor progression, supporting the biological relevance of the selected features.

Figure 4. Heatmap showing the expression patterns of the top selected genes across different breast cancer subtypes.

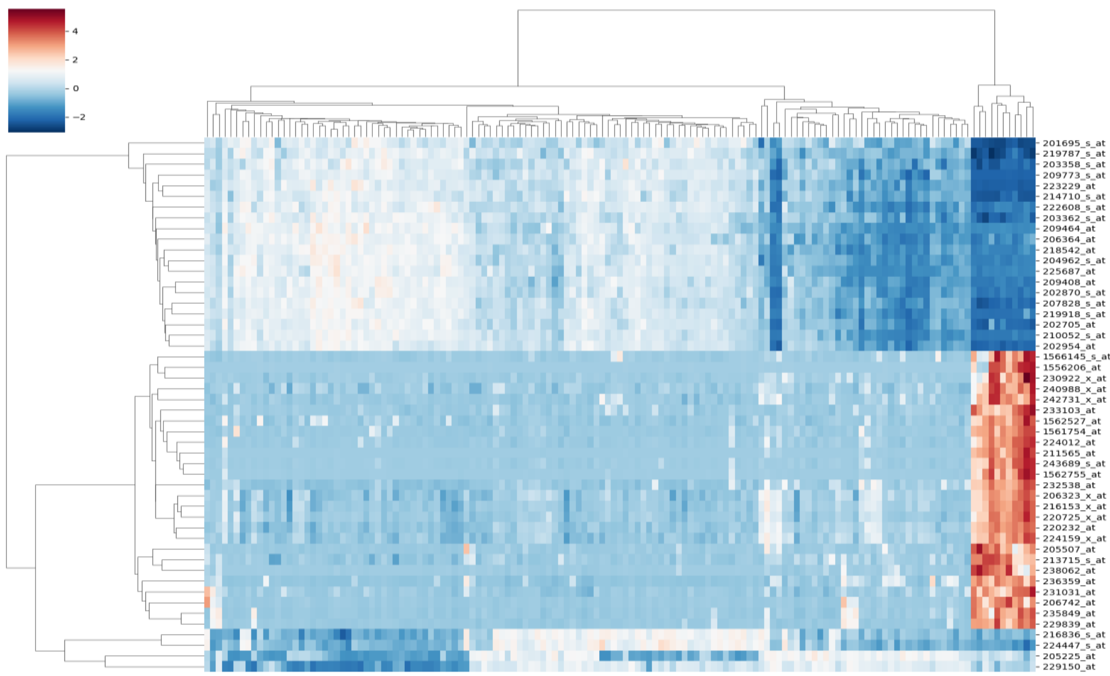
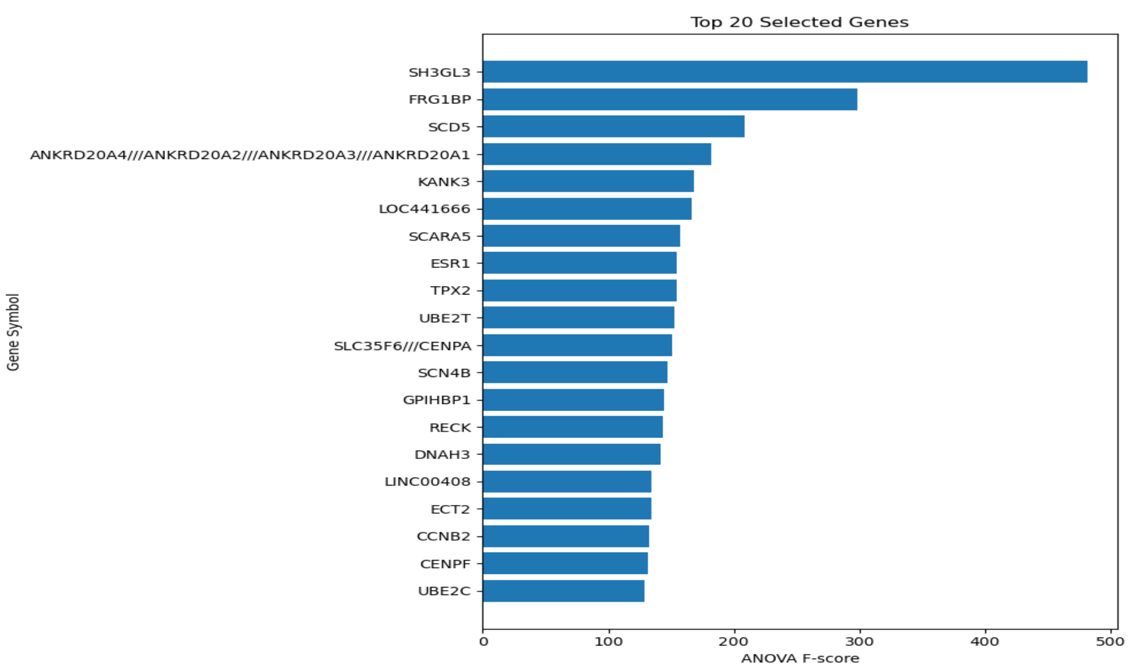


Figure 5. Top 20 ranked genes based on ANOVA F-score.



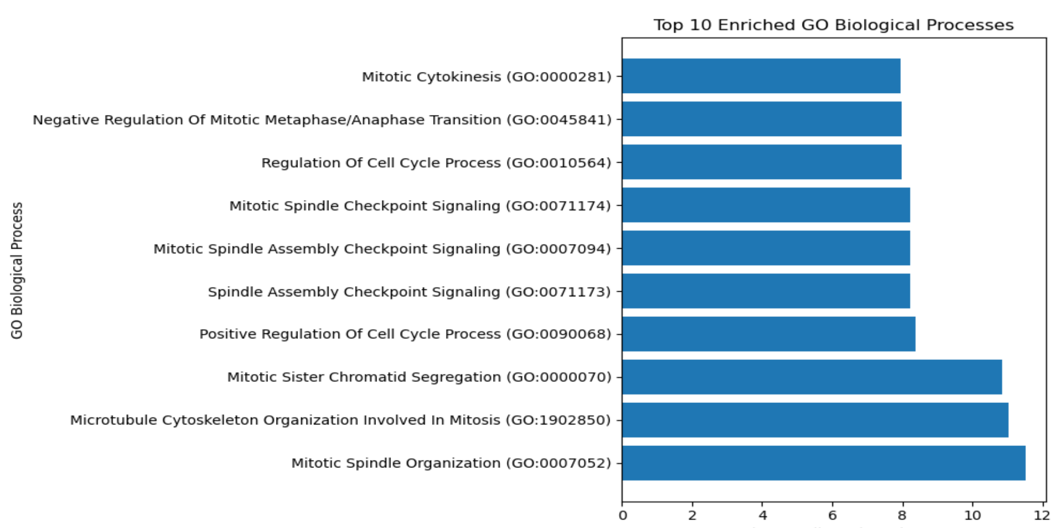
Gene Ontology (GO) Enrichment Analysis

Gene Ontology (GO) enrichment analysis was carried out to understand the biological functions of the genes selected during feature selection. The analysis showed that most of the enriched biological processes were related to cell cycle regulation and mitosis, suggesting that these genes play an important role in cell division and the growth of breast cancer cells.

The most significantly enriched GO term was Mitotic Spindle Organization, followed by Microtubule Cytoskeleton Organization Involved in Mitosis and Mitotic Sister Chromatid Segregation. Other important biological processes included Positive Regulation of Cell Cycle Process, Spindle Assembly Checkpoint Signaling, Mitotic Spindle Checkpoint Signaling, Regulation of Cell Cycle Process, Negative Regulation of Mitotic Metaphase/Anaphase Transition, and Mitotic Cytokinesis.

These processes are essential for ensuring that cells divide correctly and maintain the proper distribution of chromosomes during cell division. When these mechanisms become disrupted, uncontrolled cell growth and genomic instability can occur, which are well-known characteristics of cancer. Therefore, the GO enrichment results support the biological importance of the genes identified in this study and indicate that they are closely associated with breast cancer progression.

Figure 6. Top enriched Gene Ontology (GO) Biological Process terms.



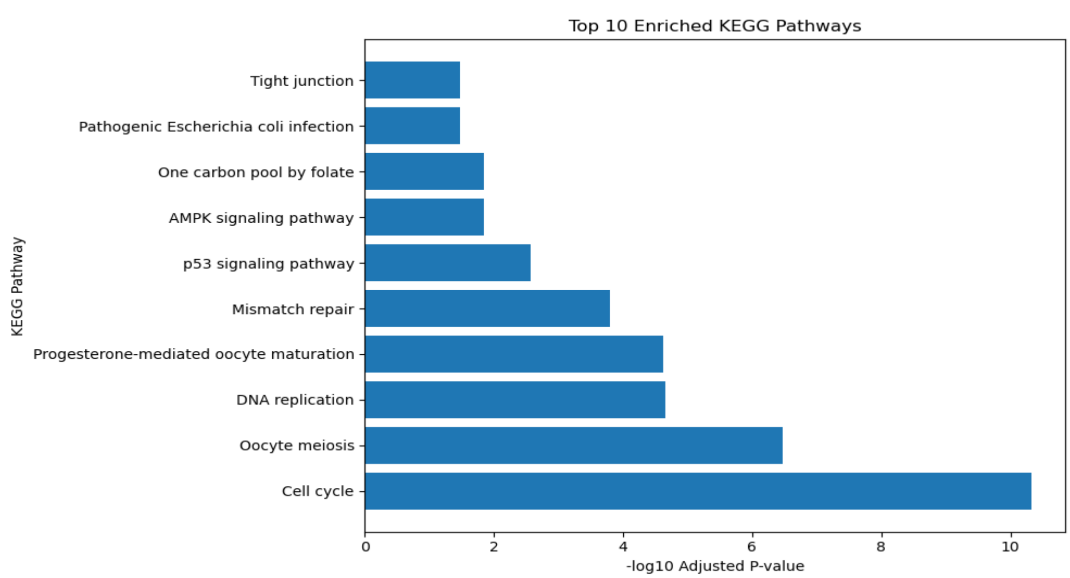
KEGG Pathway Enrichment Analysis

KEGG pathway enrichment analysis was performed to identify the biological pathways in which the selected genes are involved. The results showed that the genes were significantly enriched in several pathways that are known to be associated with cell growth, DNA maintenance, and cancer development.

Among all the pathways, the Cell Cycle pathway showed the highest level of enrichment. Other significantly enriched pathways included Oocyte Meiosis, DNA Replication, Progesterone-Mediated Oocyte Maturation, Mismatch Repair, and the p53 Signaling Pathway. Additional pathways such as the AMPK Signaling Pathway, One Carbon Pool by Folate, Pathogenic *Escherichia coli* Infection, and Tight Junction were also identified.

Many of these pathways are directly involved in regulating cell division, repairing DNA damage, and maintaining normal cellular function. In particular, the enrichment of the Cell Cycle, DNA Replication, Mismatch Repair, and p53 Signaling pathways highlights their important role in breast cancer biology. These findings suggest that the genes selected by the machine learning models are not only useful for classification but are also biologically relevant and involved in key mechanisms underlying breast cancer.

Figure 7. Top enriched KEGG pathways identified from the selected genes.



Overall, both the GO and KEGG enrichment analyses consistently indicated that the selected genes are mainly involved in cell cycle regulation, mitotic processes, DNA replication, and DNA repair. These findings support the machine learning results and demonstrate that the identified genes are biologically relevant to breast cancer molecular subtype classification.

Discussion

The main objective of this study was to classify breast cancer molecular subtypes using gene expression data and to compare the performance of different machine learning algorithms. The results showed that all four models were able to classify the breast cancer subtypes with high accuracy, demonstrating the usefulness of machine learning for analyzing high-dimensional gene expression data.

Among the four models, Logistic Regression and Support Vector Machine (SVM) achieved the highest test accuracy of 100%. However, relying only on test accuracy can sometimes give an overly optimistic estimate of model performance, especially when working with relatively small datasets. To obtain a more reliable evaluation, 10-fold stratified cross-validation was also performed. The cross-validation results showed that SVM achieved the highest mean accuracy (96.44%) with the lowest standard deviation, indicating that it provided the most stable and consistent performance across different data splits. Logistic Regression also showed strong performance, while Random Forest and XGBoost achieved slightly lower but still competitive accuracies.

One of the important steps in this study was feature selection using the ANOVA F-test. Gene expression datasets contain thousands of genes, many of which do not contribute significantly to disease classification. Selecting only the most informative genes reduced the dimensionality of the dataset and helped improve the efficiency of the machine learning models. More importantly, it allowed the study to focus on genes that are biologically relevant to breast cancer rather than using all available features.

To better understand the biological significance of the selected genes, Gene Ontology (GO) and KEGG pathway enrichment analyses were performed. The GO analysis showed that the selected genes were mainly involved in biological processes related to mitosis and cell cycle regulation, including mitotic spindle organization, chromosome segregation, and cytokinesis. Similarly, the KEGG analysis identified significant enrichment in pathways such as the Cell Cycle, DNA Replication, Mismatch Repair, and p53 Signaling Pathway. These pathways are well known for their roles in regulating cell growth, maintaining genomic stability, and preventing uncontrolled cell division. The consistency between the GO and KEGG analyses strengthens the biological relevance of the genes identified during feature selection.

The findings of this study are consistent with previous research showing that machine learning algorithms can successfully classify breast cancer molecular subtypes using gene expression data. However, unlike many studies that focus only on prediction accuracy, this work also explored the biological functions and pathways associated with the selected genes. Combining predictive modeling with functional enrichment analysis provides a more comprehensive understanding of the molecular mechanisms involved in breast cancer.

Despite these encouraging results, this study has some limitations. The analysis was performed using a single publicly available microarray dataset containing 141 samples after preprocessing. Although cross-validation was used to improve the reliability of the evaluation, testing the models on independent external datasets would provide stronger evidence of their generalizability. In addition, only microarray gene expression data were analyzed. Future studies could include RNA sequencing data, larger datasets, or multi-omics data to improve predictive performance and biological interpretation.

Overall, the results demonstrate that machine learning, together with statistical feature selection and functional enrichment analysis, can effectively classify breast cancer molecular subtypes while also providing meaningful biological insights. These findings highlight the potential of computational approaches to support breast cancer research and contribute to the development of precision medicine.

Conclusion

This study demonstrated the application of machine learning techniques for the classification of breast cancer molecular subtypes using gene expression data from the GSE45827 dataset. After preprocessing the data and selecting the most informative genes using the ANOVA F-test, four machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were trained and evaluated.

Among the evaluated models, Logistic Regression and SVM achieved the highest test accuracy of 100%, while cross-validation results showed that SVM provided the most consistent overall performance. These findings indicate that machine learning models are capable of accurately distinguishing between different breast cancer molecular subtypes using gene expression profiles.

The biological interpretation of the selected genes further strengthened the study. GO enrichment analysis revealed that many of the identified genes are involved in cell cycle regulation and mitotic processes, while KEGG pathway analysis highlighted important pathways such as the Cell Cycle, DNA Replication, Mismatch Repair, and p53 Signaling Pathway. These findings are consistent with the known molecular mechanisms involved in breast cancer development and progression.

Although the study was conducted using a relatively small publicly available dataset, the combination of feature selection, cross-validation, and functional enrichment analysis improved the reliability and biological relevance of the results. Future work may include validation using independent datasets, RNA sequencing data, and deep learning approaches to further improve prediction performance.

In conclusion, this study shows that integrating machine learning with biological pathway analysis provides an effective approach for breast cancer molecular subtype classification. The findings contribute to a better understanding of breast cancer biology and demonstrate the potential of computational methods to support cancer research and future precision medicine applications.

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